SAFESITE

Construction Workplace Risk Assessment

Powered by machine learning to predict construction accident risk and recommend preventive actions

GROUP 1 PROJECT

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CONSTRUCTION SAFETY | DATA | MACHINE LEARNING

Dataset & Data Structure

4,847
Incident Records

29
Columns/Variables

10 +19
Integer + text variables

OSHA (Occupational Safety and Health Administration) Construction Incident Dataset) (2015-2017)

The dataset contains historical construction workplace incident records collected between 2015 and 2017. It provides information on the types of incidents that occurred, as well as relevant environmental, human, and task-related factors. These records were analyzed to identify patterns in construction workplace incidents and provide the information needed to develop SafeSite's machine learning model for assessing the risk level of future incident profiles.

PROBLEM DEFINITION

Construction is one of the most hazardous industries worldwide. The International Labor Organization (ILO) estimates that around 60,000 fatal accidents occur on construction sites globally each year, with approximately one in six fatal workplace accidents occurring in construction.

Beyond the number of incidents, these events are influenced by environmental, human and task-related factors.

The problem we identified is that historical incident data contains valuable patterns, but these patterns are not always translated into an assessment of the risk associated with a particular incident profile.

SAFESITE was developed to use these historical patterns to classify an incident as (Low/Medium/High) and provide a corresponding safety action.

01 WHY IT MATTERS

- 60,000

Fatal construction accidents occur globally each year.

Construction workers face significantly higher risks of fatal workplace accidents compared to many other occupations

02 THE PROBLEM (THE GAP)

Historical incident records contain valuable information, but identifying which combinations of incident type, environmental conditions, human factors and task characteristics represent higher risk is time-consuming and not always accurate.

03 OUR RESPONSE

SafeSite

DATA • INSIGHT • SAFETY

MODEL PURPOSE & SELECTION

How we used the data to identify incident risk

1. WHAT WAS THE MODEL DESIGNED TO DO?

The model was developed to learn from historical construction incident records and estimate whether an incident profile was likely to result in a FATAL or NON-FATAL outcome.

In safe site, this predicted fatality probability is translated into Low, Medium or High Risk to make the result easier to interpret and act upon

2. WHY IS CLASSIFICATION APPROPRIATE

CATEGORICAL OUTCOME

The original outcome has two categories:

- Fatal
- Non-Fatal

Therefore, this is a classification problem, not a regression problem.

MULTIPLE FACTORS

The incident outcome is not determined by one variable alone. The dataset contains several characteristics:

- Event type
- Environmental Factor
- Human Factor & Tasks

3. WHY DID WE COMPARE SEVERAL MODELS?

Rather than assuming that one algorithm would be best, we trained and evaluated several classification models and compared their performance

LOGISTIC REGRESSION

VS

NAÏVE BAYES

VS

GRADIENT BOOSTING

VS

RANDOM FOREST

4. WHY RANDOM FOREST?

Random forest is well suited to tabular data containing multiple categorical and interacting factors. It combines multiple decision trees to make a final prediction, allowing it to capture more complex relationships incident characteristics than a single decision tree

MODEL PERFORMANCE

ACCURACY 67.1%	FATAL RECALL 72%
MACRO F1 0.656	NON-FATAL RECALL 59%

05 • EXPECTED OUTCOME

What we expected before training

Before training, the team expected that a classifier built on the four available incident features — event type, environmental factor, human factor, and task assignment — could learn enough signal from 4,847 historical records to separate fatal from non-fatal outcomes well enough to be operationally useful, even without medical or biometric data.

1

Learnable pattern

We expected certain event/environment/human-factor combinations to correlate strongly enough with fatal outcomes for a model to pick up on.

2

Above-chance accuracy

With a near-even class split (61% fatal / 39% non-fatal), we expected the model to clearly beat a naive baseline guess.

3

Usable risk tiers

We expected predicted probabilities could be grouped into Low / Medium / High risk bands meaningful enough to guide action.

4

A deployable tool

We expected the trained model could be wrapped in a simple interface an HSE officer could use without any data science background.

Architecture, workflow and the deployed app

SafeSite is deployed as a Streamlit web app, turning the trained Random Forest model into a tool any HSE officer can use directly in the browser — no code, no notebook.

User input (4-step form) → Encoders → **Random Forest model** → Probability output → Risk level (Low/Med/High) → Safety action recommendation

Environmental and human-factor dropdowns are context-sensitive — narrowed to fit the selected incident type before reaching the model.

- **Three-page navigation**

  Overview, Incident Insights and Assess an Incident are one click apart via the sidebar.

- **Cached loading**

  Model, encoders and dataset load once (@st.cache_resource / @st.cache_data) so the app stays fast.

- **Guided, context-aware form**

  Assess an Incident walks the user through 4 steps, narrowing choices to fit the selected incident type.

Overview — live dataset metrics on load

Where the risk concentrates

4,847	Total incidents
2,964	Fatal incidents
1,883	Non-fatal incidents
61.2%	Fatal share

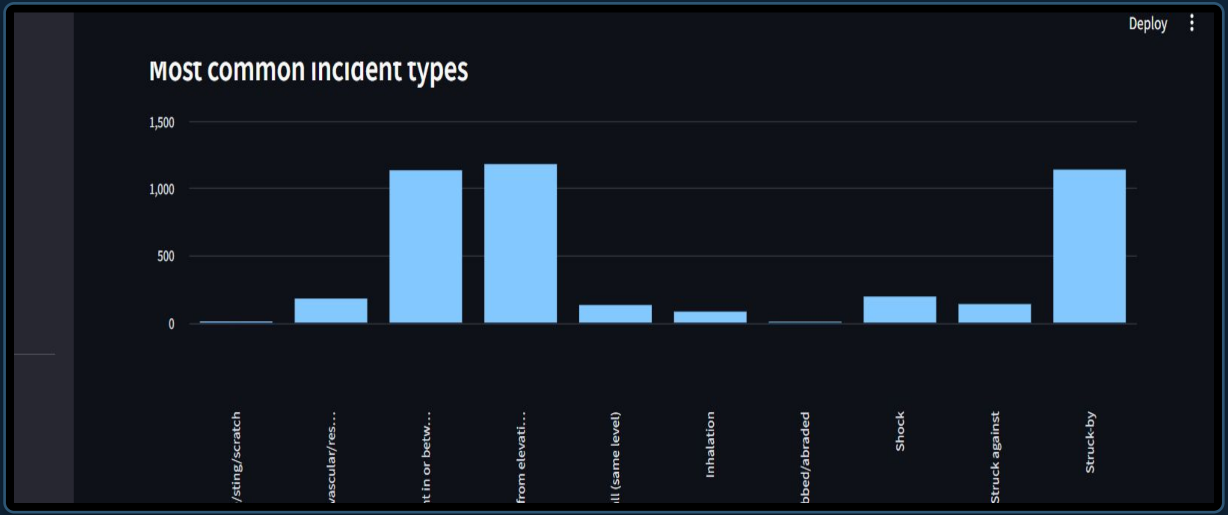
MODEL PERFORMANCE

67.1%	Accuracy
0.656	Macro F1
72%	Fatal recall
59%	Non-fatal recall

Confusion matrix (test set): 427 fatal correctly identified, 223 non-fatal correctly identified — 153 fatal cases missed, 165 non-fatal cases misclassified as fatal.

Highest fatal-share incident types

Incident type	Incidents	Fatal share
Ingestion	5	100.0%
Card-vascular / resp. fail.	179	98.3%
Shock	194	83.5%
Struck against	138	58.0%
Bite / sting / scratch	9	44.4%
Fall (same level)	131	39.7%
Caught in or between	1,133	37.6%

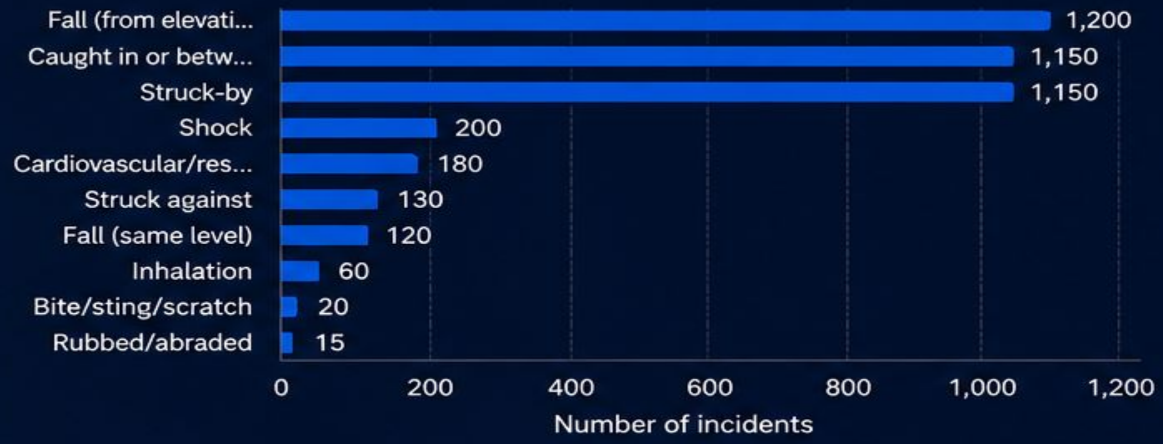


"Caught in or between" is most common by volume (1,133 cases); ingestion, cardio-vascular failure and shock carry the highest fatal share.

Key findings & what they mean

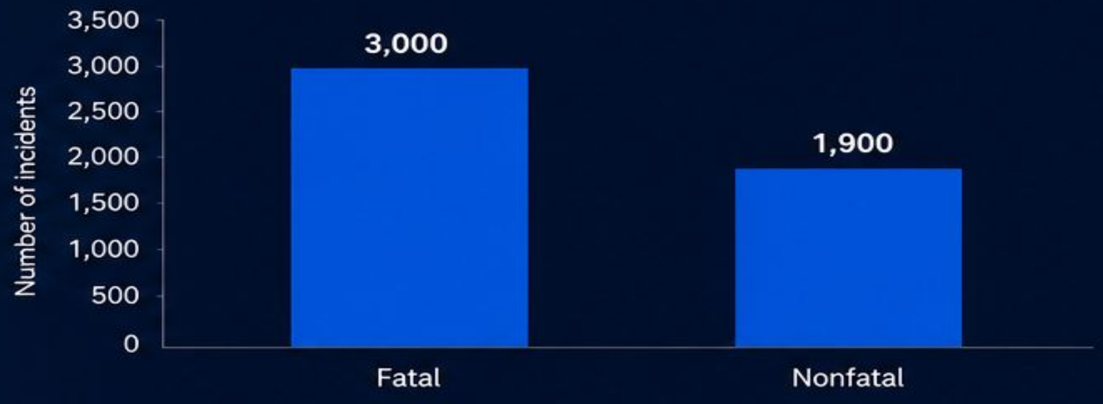
Incident patterns

Most common incident types in the dataset.



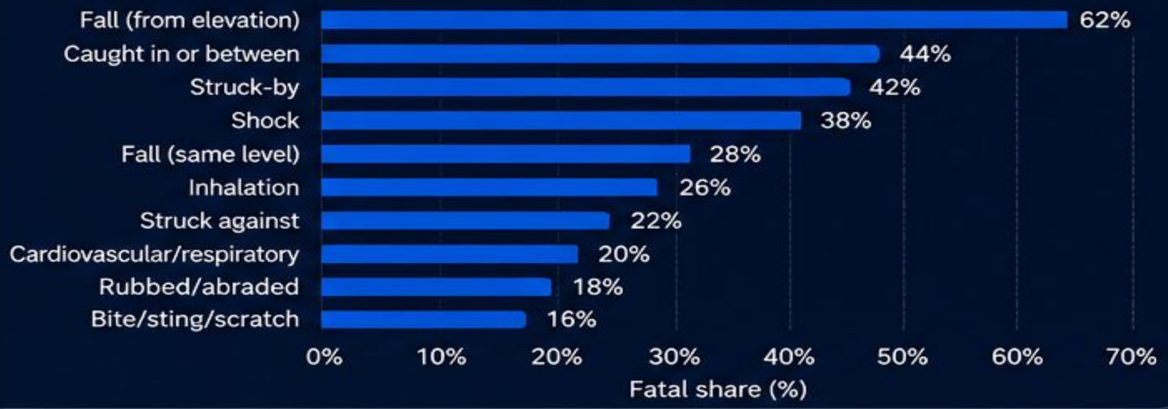
Outcome pattern

Fatal vs nonfatal distribution in the dataset.



Highest-risk profile

Incident types ranked by fatal share in the dataset.



Model performance

Performance of the selected Random Forest model.

67.1%	0.656	72%	59%
Accuracy	Macro F1	Fatal Recall	Nonfatal Recall

Model selected based on highest Macro F1 to balance performance across Fatal and Nonfatal classes.

The goal is not only to classify incidents — it is to use historical evidence to support better preventive decisions.

Recommendations & conclusion

01 Prioritise high-risk incident profiles

Prioritise immediate hazard review and strengthen controls before the task proceeds. Review controls before similar tasks continue.

02 Strengthen task-specific safety

For **Medium-risk profiles**, review the selected hazards and verify that effective controls are in place before and during the task Reinforce training, supervision and safe procedures.

03 Improve protective controls

Ensure appropriate **hazard controls and preventive measures** are consistently reviewed and maintained for the identified task conditions. Ensure appropriate Personal Protective Equipment and hazard controls are consistently used.

04 Keep learning from incidents

Continue monitoring incident data and **retrain/re-evaluate the model as new records become available**. Continue monitoring incident data and update the model as new records become available.

SafeSite turns historical construction incident data into practical information for safer workplace decisions by combining Fatal/Nonfatal outcome prediction with Low, Medium and High operational risk levels.

QUESTIONS?

Summary & Key Takeaways